

1. The algorithms utilized in generating hash values are:

(1) MD5 (2) SHA-1 (3) SHA-256 (4) RSA (5) AES (6) DES

Answer: (1) + (2) + (3)

2. It's not an active type attack:

- a) The attacker captures messages and sends them forward
- b) The attacker reads messages sent in clear on the communications channel**
- c) The attacker sends messages under a different identity
- d) The attacker resends messages captured in a different session
- e) Man in the middle

Answer: b)

3. Indicate which of the following statements represent characteristics (objectives) of a modern cryptographic system (more than one statement can be correct)

- a) Nonrepudiation**
- b) Implements cryptographic algorithms secretly developed by teams of specialists
- c) Assures the authentication of sources**
- d) Assures data integrity**
- e) Assures total or partial confidentiality of the message**
- f) Does not assure the authentication of sources
- g) Does not assure data integrity

Answer: a) + c) + d) + e)

4. What categories of cryptographic services does the X.800 standard define?

(1) Authentication (2) Access Control (3) Data Confidentiality (4) Data Integrity (5) Nonrepudiation

Answer: (1) + (2) + (3) + (4) + (5)

5. The hash value is used for:

- a) To prevent situations where the source does not recognize the message
- b) To guarantee the identity of all parties
- c) To guarantee the message source
- d) To guarantee the message integrity**
- e) To hide the message contents

Answer: d)

6. In a cryptographic analysis of a communication environment (for example the Dolev-Yao model) an attacker CANNOT do (two correct answers):

- a) He can become the message recipient
- b) He can send messages in the source's name
- c) He has full control over the network
- d) He can determine the public key associated to a private key
- e) He can determine the private key associated to a public key**
- f) He can gain any message sent on the network
- g) He can guess a random number from a large enough pool**

Answer: e) + g)

7. Be it the binary values $x_2=10101110$ and $y_2=11010011$ and the mathematical expression:

$z_2 = x_2 \text{ XOR } y_2$; $h_2 = z_2 \text{ XOR } y_2$; Indicate the correct values for z_2 and h_2 :

- a) $z_2 = 11111111$ and $h_2=10101110$
- b) $z_2 = 10000010$ and $h_2 = 11010011$
- c) $z_2 = 01111101$ and $h_2 = 10101110$**
- d) $z_2 = 10000010$ and $h_2 = 10101110$
- e) $z_2 = 11111111$ and $h_2 = 11010011$

Answer: c)

8. Choose the WRONG statement regarding the mathematical problems of the NP class complexity:

- a) **The problem of allocating 10000 students based on 300 criteria is an NP complexity class**
- b) The knapsack problem is NOT an NP class complexity
- c) They are the base of an unsolved mathematical theory
- d) The Diffie-Hellman problem is an NP class complexity
- e) The RSA algorithm uses algorithms from the NP class complexity

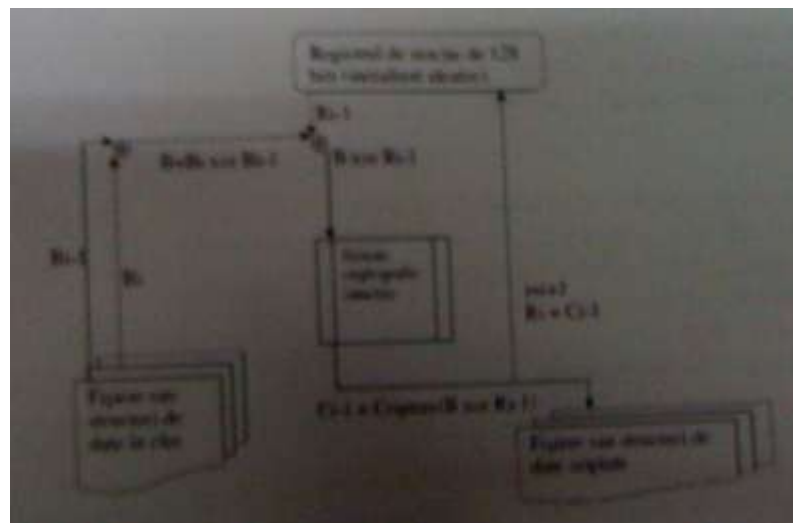
Answer: a)

9. Choose the correct statement regarding the cryptography principles

- a) Best-practice is to keep the algorithm a secret
- b) **The algorithms used have solutions that are impossible to determine**
- c) The algorithms used have solutions that can be determined in a certain/limited period of time
- d) The algorithms used DO NOT implement mathematical models of the NP class complexity
- e) The algorithms used do not require a higher computational power to check the solution

Answer: b)

10. The following diagram represents the cipher mode (any type of algorithm with a symmetrical key can be included in the diagram):



- a) ECB cipher mode – Electronic Code Book
- b) PCBC cipher mode – Propagation Cipher Block Chaining**
- c) Sequential cipher mode
- d) Block Cipher Chaining mode (CBC)
- e) None of the above

Answer: b)

11. Considering $N = 247$ and the set of integer values (1, 2, 3, ..., 189, 190). Using the knapsack problem, find all subsets of elements of the set which is a solution for N and choose which statement is a solution:

- a) {10, 20, 30, 40, 47, 101}
- b) The time allotted for the exam is NOT enough to solve the problem
- c) {1, 29, 30, 40, 47, 101}
- d) {10, 20, 30, 43, 44, 100}**
- e) {10, 25, 30, 42, 47, 99}

Answer: d)

12. It is NOT a way to generate random numbers to be used in a cryptographic algorithm:

- a) Pressing keys in an interval
- b) The cursor position in an interval
- c) Position of a ...
- d) The C rand function**
- e) The adding congruence method

Answer: d)

13. Which of the following methods is a pseudo-random number generator (PRNG):

- a) Yarrow
- b) Blum Blum shub
- c) ANSI X9.17
- d) Linear congruence model
- e) Linear feedback shift registers generator

Answer: a) + b) + c) + d) + e)